

ASHIRWAD Periodic Test – II (Grade 7 – Robotics) – ANSWER KEY

I. Answer the following Questions

1. Stepper Motor
2. 0° to 180°
3. c) Actuator
4. c) Motor uses electricity to move; generator uses motion to make electricity
5. Match the following:
 - a. DC Motor → Toy Car / Fan
 - b. Servo Motor → Robot Arm / Smart Lock
 - c. Stepper Motor → 3D Printer / CNC Machine
 - d. Generator → Wind Turbine / Bicycle Light
 - e. Motor in Washing Machine → Rotates Drum to Wash Clothes
6. b) A component that performs physical movement

II. STEM Activity

7. b) The polarity of the battery was reversed
8. b) By observing the brightness of the LED
9. a) DC Motor
 - b) Mechanical → Electrical
 - c) Spin the handle faster

III. Motor Experiment

Objective:

To make a DC motor run using a battery and understand electric-to-mechanical energy conversion.

Materials Required:

DC motor, battery, wires, switch.

Circuit Diagram:

Battery (+) → Motor → Battery (–)

Demo:

When the wires are connected to the battery, the motor starts rotating. Reversing polarity reverses the direction.

Result:

The motor converts electrical energy into mechanical energy.